Lesson 3.3 - How to Calculate the Median Using a Formula

> **Video:** How To Calculate The Median Using a Formula by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

The formulas

First **sort the data** in ascending order. Let a_k be the k-th value in the sorted list.

When n is odd:

$$\text{median} = a_{\frac{n+1}{2}}$$

When n is even:

$$\text{median} = \frac{a_{\frac{n}{2}} + a_{\frac{n}{2}+1}}{2}$$

This is the average of the two middle terms.

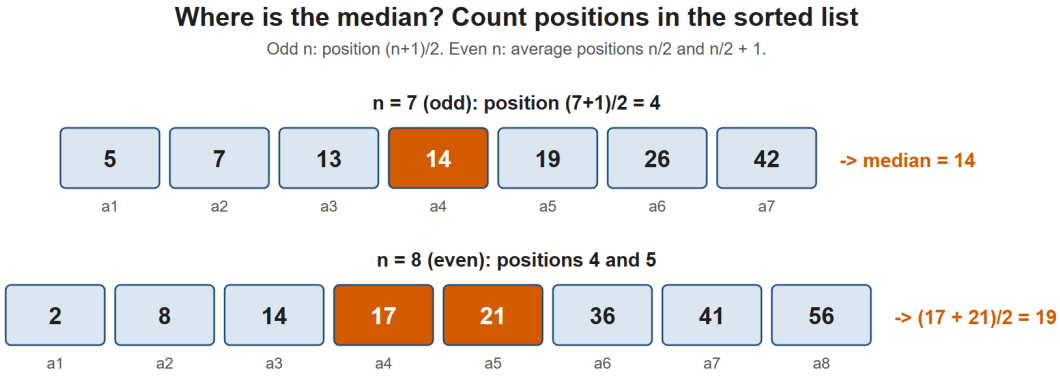
The formula tells you **which position** holds the median, which is handy when n is large.

Example 1 - odd n

Data: 7, 14, 5, 19, 26, 42, 13

Sorted: 5, 7, 13, **14**, 19, 26, 42 ($n = 7$)

$$\frac{n+1}{2} = \frac{8}{2} = 4 \;\Rightarrow\; \text{median} = a_4 = 14$$



The formula only tells you a position in the sorted list. With 7 values the 4th box is the median. With 8 values there is no single middle box, so the 4th and 5th are averaged.

Example 2 - even n

Data: 8, 21, 14, 36, 17, 2, 56, 41

Sorted: 2, 8, 14, **17, 21**, 36, 41, 56 ($n = 8$)

$$\frac{n}{2} = 4, \quad \frac{n}{2} + 1 = 5 \;\Rightarrow\; \text{median} = \frac{a_4 + a_5}{2} = \frac{17 + 21}{2} = 19$$

Key takeaways

- Always sort first. The formula gives a **position**, not a value.
- Odd n: one middle term at position $(n + 1)/2$.
- Even n: average the terms at positions $n/2$ and $n/2 + 1$.